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Cybersecurity Technical Assessment Report

This report summarizes the technical implementations and security audit findings based on two automated scripts developed for network protocol poisoning and mobile device vulnerability scanning.

1. Active Directory Protocol Poisoning Audit

Target Protocol: LLMNR (Link-Local Multicast Name Resolution) and NBT-NS (NetBIOS Name Service).

Tooling: NBT_final.py (Python with Scapy).

Technical Overview

The script NBT_final.py facilitates a passive-to-active attack vector on internal networks by sniffing and spoofing broadcast name resolution requests.

- **Traffic Interception:** The script monitors **UDP ports 137 (NetBIOS) and 5355 (LLMNR)**. It uses the Scapy sniff function to detect name queries that occur when a Windows host cannot resolve a hostname via DNS.
- **Packet Spoofing:** Upon detecting a QUERY payload, the script constructs a spoofed response packet. This redirects the victim's authentication attempt to an auditor-controlled IP by sending a raw FAKE_RESPONSE payload.
- **Credential Capture:** The logic includes a capture_credentials function designed to log intercepted authentication attempts into hospital_hashes.txt for offline cryptographic analysis.

Security Impact

If LLMNR/NBT-NS protocols are enabled, an attacker can intercept NTLM hashes. These hashes can be subjected to brute-force attacks or utilized in "Pass-the-Hash" maneuvers to gain unauthorized access to network resources.

2. Android Mobile Vulnerability Assessment

Target Platform: Android OS via ADB (Android Debug Bridge).

Tooling: mobile_final.py (Python with Subprocess).

Technical Overview

The mobile_final.py script provides a modular framework for auditing the security posture of mobile endpoints connected via USB debugging.

- **Integrity Check (Root Detection):** The script executes the `id` command through a superuser (`su`) shell. If the returned UID is 0, the device is flagged as **ROOTED**, indicating that the Android security sandbox has been compromised.
- **Network Attack Surface:** It utilizes the `netstat` command to inventory open network ports. Exposed ports on a mobile device can indicate unauthorized remote access points or vulnerable listening services.
- **Application Audit:** The script interacts with the Android package manager (`dumpsys package`) to retrieve `versionCode` data. This allows analysts to identify outdated third-party applications that may contain known vulnerabilities.

Security Impact

Rooted devices and exposed ports significantly increase the risk of data exfiltration and malware persistence. This automated scan allows for rapid identification of high-risk endpoints within a corporate or clinical environment.